DANIEL MCKNIGHT

0410 225 895 | daniel.mcknight@dpird.nsw.gov.au | linkedin.com/in/daniel-mcknight-59a742167

PROFESSIONAL PROFILE

Highly driven Bioinformatician with a recently submitted PhD thesis and a strong record in developing automated, reproducible analysis pipelines (Bash/Python/R). Proven ability to directly apply advanced genomics, phylogenetics, and pangenomics to biosecurity and molecular diagnostics, including hands-on experience in high-throughput NATA-accredited laboratories. Adept at collaborative research, having successfully coordinated projects across multiple institutes, and committed to advancing open-source science and rapidly acquiring new technical skills to deliver timely and accurate results. I am eager to utilise the skills from my PhD to continue making tangible contributions to Australian biosecurity.

EMPLOYMENT

2021 to present	Technical assistant: Biotechnology and Plants laboratories at the NSW Department of Primary Industries and Regional Development (DPIRD)
2020 to 2021	Electrical contractor: Devika
2020	Summer Casual Researcher: Yu Research Group
2018	Biochemistry intern: Venus Shell Systems
2015 to 2020	Sales Assistant: Jaycar Electronics, Wollongong

EDUCATION

2021 to 2025	Doctor of Philosophy, a collaborative project between the University of Technology Sydney (UTS) and the DPIRD
2020	First-class Honours at the University of Wollongong
2016 to 2019	Bachelor of Science (Biological Sciences) at the University of Wollongong

TECHNICAL SKILLS

- **Proficient at developing and validating qPCR primers:** I developed these skills to create an assay for pre-symptomatic detection of *Xanthomonas campestri*s in broccoli.
- **High digital literacy:** experience in Python, R, and GNU bash from 5 years of bioinformatics/computational biophysics research.
- **Data analysis:** experience in data manipulation and visualisation from my bioinformatics PhD research for my published papers and thesis
- **Adept at bioinformatic techniques**: including phylogenetics, genomic/pangenomic analysis, bacterial genome assembly, virulence factor/genomic island identification, data manipulation, and scripting.

SOFT SKILLS

- **Communication and collaboration:** Effectively coordinated research across 12 co-authors from four different institutes, demonstrating adept communication and the ability to form productive relationships with diverse interdisciplinary teams.
- **Adaptability and rapid skill acquisition:** I transitioned from a biological science background to computational biophysics and bioinformatics, successfully teaching myself theory and programs to achieve first-class honours and a PhD.
- **Pragmatic problem-solving and innovation:** Developed and implemented automated bioinformatics pipelines (bash/Python) with QC checkpoints, error handling, and reproducibility to solve complex analytical challenges in genomics.

CERTIFICATES AND SCHOLARSHIPS

2024	Store and handle dangerous goods and hazardous substances (TLID0021)
2023	Vice-Chancellor's Conference Fund at UTS
2021	Australian Research Council (ARC) linkage scholarship and stipend

COMMUNITY INVOLVEMENT

2023 to present	Volunteer at Illawarra Potters Inc., a community pottery studio
2020	Student representative for SMAH honours course, UOW
2019	Learning Labs, UOW - Volunteer to teach high school students in Summer STEM classes
2018 to 2019	BioNet, UOW - A social and support network for students studying biology-based degrees

REFEREES

Dr Daniel Bogema
Senior Research Scientist
PhD Supervisor
daniel.bogema@dpi.nsw.gov.au
02 4640 6154

Dr Toni Chapman
Research Scientist
PhD supervisor
toni.chapman@dpi.nsw.gov.au

02 4640 6219